LAB Assignment

(Studio 3T)

PRATIKSHA PAL(AIML 064)

COLLECTION 1: **flights**

Schema

{

flight_id: Number,

airline: String,

source: String,

destination: String,

price: Number,

duration: Number

}

Table

flight_id	airline	source	destination	price	duration
1	Indigo	Delhi	Mumbai	5000	2
2	AirIndia	Kolkata	Chennai	6500	3
3	SpiceJet	Mumbai	Bangalore	4000	1.5
4	Vistara	Delhi	Chennai	7000	3
5	GoFirst	Mumbai	Delhi	4800	2
6	Indigo	Bangalore	Kolkata	5500	2.5

COLLECTION 2: **bookings**

Schema

{

booking_id: Number,

passenger_name: String,

flight_id: Number,

city: String,

seats: Number,

total_fare: Number

}

Table

booking_id	passenger_name	flight_id	city	seats	total_fare
101	Aarav	1	Delhi	1	5000
102	Meera	2	Kolkata	2	13000
103	Rohan	3	Mumbai	1	4000
104	Sneha	4	Delhi	3	21000
105	Kunal	5	Mumbai	2	9600
106	Priya	6	Bangalore	1	5500

QUESTIONS

Q1. Display all flights.

```
use airlineDBAIML064
db.flightsAIML64.insertMany([
  { flight_id: 1, airline: "Indigo", source: "Delhi", destination: "Mumbai",
  { flight_id: 2, airline: "AirIndia", source: "Kolkata", destination: "Cher
  { flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Banga
  { flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chennai
  { flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Delhi"
  { flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "Kolk
])
db.bookingsAIML64.insertMany([
  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", s
  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata",
  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai",
  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", s
  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai",
  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore
])
db.flightsAIML64.find()
```

Raw shell output | Shell Output (Documents) | Find Query (line 18)

50 Documents 1 to 6

```
{
    "_id" : ObjectId("69d0159b1293150be244152e"),
    "flight_id" : NumberInt(1),
    "airline" : "Indigo",
    "source" : "Delhi",
    "destination" : "Mumbai",
    "price" : NumberInt(5000),
    "duration" : NumberInt(2)
}
{
    "_id" : ObjectId("69d0159b1293150be244152f"),
    "flight_id" : NumberInt(2),
    "airline" : "AirIndia",
    "source" : "Kolkata",
    "destination" : "Chennai",
    "price" : NumberInt(6500),
    "duration" : NumberInt(3)
}
{
    "_id" : ObjectId("69d0159b1293150be2441530"),
    "flight_id" : NumberInt(3),
    "airline" : "SpiceJet",
    "source" : "Mumbai",
    "destination" : "Bangalore",
    "price" : NumberInt(4000),
    "duration" : 1.5
}
```

Q2. Display passenger_name and city from bookings.

```
use airlineDBAIML064
db.flightsAIML64.insertMany([
  { flight_id: 1, airline: "Indigo", source: "Delhi", destination: "Mumbai",
  { flight_id: 2, airline: "AirIndia", source: "Kolkata", destination: "Cher
  { flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Banga
  { flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chennai
  { flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Delhi'
  { flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "Kolk
])
db.bookingsAIML64.insertMany([
  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", s
  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata",
  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai",
  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", s
  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai",
  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore
])
db.flightsAIML64.find()
db.bookingsAIML64.find({}, { passenger_name: 1, city: 1, _id: 0 })
```

Raw shell output | Shell Output (Doc... | Find Query (line ... | Find Que

10 | Documents 1 to 10

```
{
    "passenger_name" : "Aarav",
    "city" : "Delhi"
}
{
    "passenger_name" : "Meera",
    "city" : "Kolkata"
}
{
    "passenger_name" : "Rohan",
    "city" : "Mumbai"
}
{
    "passenger_name" : "Sneha",
    "city" : "Delhi"
}
{
    "passenger_name" : "Kunal",
    "city" : "Mumbai"
}
{
    "passenger_name" : "Priya",
    "city" : "Bangalore"
}
{
    "passenger_name" : "Aarav",
    "city" : "Delhi"
}
```

Q3. Find flights where source is Delhi.

```
db.flightsAIML64.insertMany([
  { flight_id: 1, airline: "Indigo", source: "Delhi", destination: "Mumbai",
  { flight_id: 2, airline: "AirIndia", source: "Kolkata", destination: "Cher
  { flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Banga
  { flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chennai
  { flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Delhi'
  { flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "Kolk
])
db.bookingsAIML64.insertMany([
  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", s
  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata",
  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai",
  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", s
  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai",
  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore
])
db.flightsAIML64.find()
db.bookingsAIML64.find({}, { passenger_name: 1, city: 1, _id: 0 })
db.flightsAIML64.find({ source: "Delhi" })
```

Shell Output (... | Find Query (li... | Find Query (li... | Find Query (li...

50 | Documents 1 to 12

```
{
    "_id" : ObjectId("69d0159b1293150be244152e"),
    "flight_id" : NumberInt(1),
    "airline" : "Indigo",
    "source" : "Delhi",
    "destination" : "Mumbai",
    "price" : NumberInt(5000),
    "duration" : NumberInt(2)
}
{
    "_id" : ObjectId("69d0159b1293150be2441531"),
    "flight_id" : NumberInt(4),
    "airline" : "Vistara",
    "source" : "Delhi",
    "destination" : "Chennai",
    "price" : NumberInt(7000),
    "duration" : NumberInt(3)
}
{
    "_id" : ObjectId("69d017941293150be244153a"),
    "flight_id" : NumberInt(1),
    "airline" : "Indigo",
    "source" : "Delhi",
    "destination" : "Mumbai",
    "price" : NumberInt(5000),
    "duration" : NumberInt(2)
}
{
    "_id" : ObjectId("69d017941293150be244153d"),
    "flight_id" : NumberInt(4),
    "airline" : "Vistara",
    "source" : "Delhi",
```

Q4. Find bookings where seats = 1.

```mongo
use airlineDBAIML064
db.flightsAIML64.insertMany([
  { flight_id: 1, airline: "Indigo", source: "Delhi", destination: "Mumbai",
  { flight_id: 2, airline: "AirIndia", source: "Kolkata", destination: "Cher
  { flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Bange
  { flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chennai
  { flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Delhi'
  { flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "Kolk
])
db.bookingsAIML64.insertMany([
  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", s
  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata",
  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai",
  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", s
  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai",
  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore
])
db.flightsAIML64.find()
db.bookingsAIML64.find({}, { passenger_name: 1, city: 1, _id: 0 })
db.flightsAIML64.find({ source: "Delhi" })
db.bookingsAIML64.find({ seats: 1 })
```

Find Query (li... | Find Query (li... | Find Query (li... | Find Query (li...

50 | Documents 1 to 21


```
{
    "_id" : ObjectId("69d0159b1293150be2441534"),
    "booking_id" : NumberInt(101),
    "passenger_name" : "Aarav",
    "flight_id" : NumberInt(1),
    "city" : "Delhi",
    "seats" : NumberInt(1),
    "total_fare" : NumberInt(5000)
}
{
    "_id" : ObjectId("69d0159b1293150be2441536"),
    "booking_id" : NumberInt(103),
    "passenger_name" : "Rohan",
    "flight_id" : NumberInt(3),
    "city" : "Mumbai",
    "seats" : NumberInt(1),
    "total_fare" : NumberInt(4000)
}
{
    "_id" : ObjectId("69d0159b1293150be2441539"),
    "booking_id" : NumberInt(106),
    "passenger_name" : "Priya",
    "flight_id" : NumberInt(6),
    "city" : "Bangalore",
    "seats" : NumberInt(1),
    "total_fare" : NumberInt(5500)
}
{
    "_id" : ObjectId("69d017941293150be2441540"),
    "booking_id" : NumberInt(101),
    "passenger_name" : "Aarav",
    "flight_id" : NumberInt(1),
```


Q5. Sort flights by price in ascending order.

```mongo
use airlineDBAIML064
db.flightsAIML64.insertMany([
  { flight_id: 1, airline: "Indigo", source: "Delhi", destination: "Mumbai",
  { flight_id: 2, airline: "AirIndia", source: "Kolkata", destination: "Cher
  { flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Bange
  { flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chennai
  { flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Delhi'
  { flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "Kolk
])
db.bookingsAIML64.insertMany([
  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", s
  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata",
  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai",
  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", s
  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai",
  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore
])
db.flightsAIML64.find()
db.bookingsAIML64.find({}, { passenger_name: 1, city: 1, _id: 0 })
db.flightsAIML64.find({ source: "Delhi" })
db.bookingsAIML64.find({ seats: 1 })
db.flightsAIML64.find().sort({ price: 1 })
```

Find Query (li... | Find Query (li... | Find Query (li... | Find Query (li...

50 | Documents 1 to 48


```
{
    "_id" : ObjectId("69d0159b1293150be2441530"),
    "flight_id" : NumberInt(3),
    "airline" : "SpiceJet",
    "source" : "Mumbai",
    "destination" : "Bangalore",
    "price" : NumberInt(4000),
    "duration" : 1.5
}
{
    "_id" : ObjectId("69d0d2601293150be2441560"),
    "flight_id" : NumberInt(3),
    "airline" : "SpiceJet",
    "source" : "Mumbai",
    "destination" : "Bangalore",
    "price" : NumberInt(4000),
    "duration" : 1.5
}
{
    "_id" : ObjectId("69d0d28d1293150be244156c"),
    "flight_id" : NumberInt(3),
    "airline" : "SpiceJet",
    "source" : "Mumbai",
    "destination" : "Bangalore",
    "price" : NumberInt(4000),
    "duration" : 1.5
}
{
    "_id" : ObjectId("69d0d24b1293150be2441554"),
    "flight_id" : NumberInt(3),
    "airline" : "SpiceJet",
    "source" : "Mumbai",
```


Q6. Display top 2 bookings with highest total_fare.

```
use airlineDBAIML064
db.flightsAIML64.insertMany([
  { flight_id: 1, airline: "Indigo", source: "Delhi", destination: "Mumbai",
  { flight_id: 2, airline: "AirIndia", source: "Kolkata", destination: "Cher
  { flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Banga
  { flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chennai
  { flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Delhi'
  { flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "Kolk
])
db.bookingsAIML64.insertMany([
  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", s
  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata",
  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai",
  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", s
  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai",
  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore
])
db.flightsAIML64.find()
db.bookingsAIML64.find({}, { passenger_name: 1, city: 1, _id: 0 })
db.flightsAIML64.find({ source: "Delhi" })
db.bookingsAIML64.find({ seats: 1 })
db.flightsAIML64.find().sort({ price: 1 })
db.bookingsAIML64.find().sort({ total_fare: -1 }).limit(2)
```

```
Find Query (li... | Find Query (li... | Find Query (li... | Find Query (
50 | Documents 1 to 2
{
    "_id" : ObjectId("69d017941293150be2441543"),
    "booking_id" : NumberInt(104),
    "passenger_name" : "Sneha",
    "flight_id" : NumberInt(4),
    "city" : "Delhi",
    "seats" : NumberInt(3),
    "total_fare" : NumberInt(21000)
}
{
    "_id" : ObjectId("69d0159b1293150be2441537"),
    "booking_id" : NumberInt(104),
    "passenger_name" : "Sneha",
    "flight_id" : NumberInt(4),
    "city" : "Delhi",
    "seats" : NumberInt(3),
    "total_fare" : NumberInt(21000)
}
```

Q7. Find flights where price is greater than 5000.

```
use airlineDBAIML064
db.flightsAIML64.insertMany([
  { flight_id: 1, airline: "Indigo", source: "Delhi", destination: "Mumbai",
  { flight_id: 2, airline: "AirIndia", source: "Kolkata", destination: "Cher
  { flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Banga
  { flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chennai
  { flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Delhi'
  { flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "Kolk
])
db.bookingsAIML64.insertMany([
  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", s
  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata",
  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai",
  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", s
  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai",
  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore
])
db.flightsAIML64.find()
db.bookingsAIML64.find({}, { passenger_name: 1, city: 1, _id: 0 })
db.flightsAIML64.find({ source: "Delhi" })
db.bookingsAIML64.find({ seats: 1 })
db.flightsAIML64.find().sort({ price: 1 })
db.bookingsAIML64.find().sort({ total_fare: -1 }).limit(2)
db.flightsAIML64.find({ price: { $gt: 5000 } })
```

```
Find Query (li... | Find Query (li... | Find Query (li... | Find Query
50 | Documents 1 to 2
{
    "_id" : ObjectId("69d017941293150be2441543"),
    "booking_id" : NumberInt(104),
    "passenger_name" : "Sneha",
    "flight_id" : NumberInt(4),
    "city" : "Delhi",
    "seats" : NumberInt(3),
    "total_fare" : NumberInt(21000)
}
{
    "_id" : ObjectId("69d0159b1293150be2441537"),
    "booking_id" : NumberInt(104),
    "passenger_name" : "Sneha",
    "flight_id" : NumberInt(4),
    "city" : "Delhi",
    "seats" : NumberInt(3),
    "total_fare" : NumberInt(21000)
}
```

Q8. Display flights where destination is not Chennai.

```
use airlineDBAIML064
db.flightsAIML64.insertMany([
  { flight_id: 1, airline: "Indigo", source: "Delhi", destination: "Mumbai",
  { flight_id: 2, airline: "AirIndia", source: "Kolkata", destination: "Cher
  { flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Banga
  { flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chennai
  { flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Delhi'
  { flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "Kolk
])
db.bookingsAIML64.insertMany([
  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", s
  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata",
  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai",
  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", s
  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai",
  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore
])
db.flightsAIML64.find()
db.bookingsAIML64.find({}, { passenger_name: 1, city: 1, _id: 0 })
db.flightsAIML64.find({ source: "Delhi" })
db.bookingsAIML64.find({ seats: 1 })
db.flightsAIML64.find().sort({ price: 1 })
db.bookingsAIML64.find().sort({ total_fare: -1 }).limit(2)
db.flightsAIML64.find({ price: { $gt: 5000 } })
db.flightsAIML64.find({ destination: { $ne: "Chennai" } })
```

Find Query (li... | Find Query (li... | Find Query (li... | Find Query

50 | Documents 1 to 44


```
{
    "_id" : ObjectId("69d0159b1293150be244152e"),
    "flight_id" : NumberInt(1),
    "airline" : "Indigo",
    "source" : "Delhi",
    "destination" : "Mumbai",
    "price" : NumberInt(5000),
    "duration" : NumberInt(2)
}
{
    "_id" : ObjectId("69d0159b1293150be2441530"),
    "flight_id" : NumberInt(3),
    "airline" : "SpiceJet",
    "source" : "Mumbai",
    "destination" : "Bangalore",
    "price" : NumberInt(4000),
    "duration" : 1.5
}
{
    "_id" : ObjectId("69d0159b1293150be2441532"),
    "flight_id" : NumberInt(5),
    "airline" : "GoFirst",
    "source" : "Mumbai",
    "destination" : "Delhi",
    "price" : NumberInt(4800),
    "duration" : NumberInt(2)
}
{
    "_id" : ObjectId("69d0159b1293150be2441533"),
    "flight_id" : NumberInt(6),
    "airline" : "Indigo",
    "source" : "Bangalore",
```


Q9. Find bookings where seats are between 1 and 2.

```
use airlineDBAIML064
db.flightsAIML64.insertMany([
  { flight_id: 1, airline: "Indigo", source: "Delhi", destination: "Mumbai",
  { flight_id: 2, airline: "AirIndia", source: "Kolkata", destination: "Cher
  { flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Banga
  { flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chennai
  { flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Delhi'
  { flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "Kolk
])
db.bookingsAIML64.insertMany([
  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", s
  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata",
  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai",
  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", s
  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai",
  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore
])
db.flightsAIML64.find()
db.bookingsAIML64.find({}, { passenger_name: 1, city: 1, _id: 0 })
db.flightsAIML64.find({ source: "Delhi" })
db.bookingsAIML64.find({ seats: 1 })
db.flightsAIML64.find().sort({ price: 1 })
db.bookingsAIML64.find().sort({ total_fare: -1 }).limit(2)
db.flightsAIML64.find({ price: { $gt: 5000 } })
db.flightsAIML64.find({ destination: { $ne: "Chennai" } })
db.bookingsAIML64.find({ seats: { $gte: 1, $lte: 2 } })
```

Find Query (li... | Find Query (li... | Find Query (li... | Find Query

50 | Documents 1 to 50


```
{
    "_id" : ObjectId("69d0159b1293150be2441534"),
    "booking_id" : NumberInt(101),
    "passenger_name" : "Aarav",
    "flight_id" : NumberInt(1),
    "city" : "Delhi",
    "seats" : NumberInt(1),
    "total_fare" : NumberInt(5000)
}
{
    "_id" : ObjectId("69d0159b1293150be2441535"),
    "booking_id" : NumberInt(102),
    "passenger_name" : "Meera",
    "flight_id" : NumberInt(2),
    "city" : "Kolkata",
    "seats" : NumberInt(2),
    "total_fare" : NumberInt(13000)
}
{
    "_id" : ObjectId("69d0159b1293150be2441536"),
    "booking_id" : NumberInt(103),
    "passenger_name" : "Rohan",
    "flight_id" : NumberInt(3),
    "city" : "Mumbai",
    "seats" : NumberInt(1),
    "total_fare" : NumberInt(4000)
}
{
    "_id" : ObjectId("69d0159b1293150be2441538"),
    "booking_id" : NumberInt(105),
    "passenger_name" : "Kunal",
    "flight_id" : NumberInt(5),
```


Q10. Update price by +500 for flights of airline "Indigo".

```mongo
use airlineDBAIML064
db.flightsAIML64.insertMany([
  { flight_id: 1, airline: "Indigo", source: "Delhi", destination: "Mumbai",
  { flight_id: 2, airline: "AirIndia", source: "Kolkata", destination: "Cher
  { flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Banga
  { flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chennai
  { flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Delhi'
  { flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "Kolk
])
db.bookingsAIML64.insertMany([
  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", s
  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata",
  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai",
  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", s
  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai",
  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore
])
db.flightsAIML64.find()
db.bookingsAIML64.find({}, { passenger_name: 1, city: 1, _id: 0 })
db.flightsAIML64.find({ source: "Delhi" })
db.bookingsAIML64.find({ seats: 1 })
db.flightsAIML64.find().sort({ price: 1 })
db.bookingsAIML64.find().sort({ total_fare: -1 }).limit(2)
db.flightsAIML64.find({ price: { $gt: 5000 } })
db.flightsAIML64.find({ destination: { $ne: "Chennai" } })
db.bookingsAIML64.find({ seats: { $gte: 1, $lte: 2 } })
db.flightsAIML64.updateMany(
  { airline: "Indigo" },
  { $inc: { price: 500 } }
)
```

```
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{
    "acknowledged" : true,
    "insertedId" : null,
    "matchedCount" : 26.0,
    "modifiedCount" : 26.0,
    "upsertedCount" : 0.0
}
```

Q11. Delete bookings where seats are greater than 2.

```mongo
use airlineDBAIML064
db.flightsAIML64.insertMany([
  { flight_id: 1, airline: "Indigo", source: "Delhi", destination: "Mumbai",
  { flight_id: 2, airline: "AirIndia", source: "Kolkata", destination: "Cher
  { flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Banga
  { flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chennai
  { flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Delhi'
  { flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "Kolk
])
db.bookingsAIML64.insertMany([
  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", s
  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata",
  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai",
  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", s
  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai",
  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore
])
db.flightsAIML64.find()
db.bookingsAIML64.find({}, { passenger_name: 1, city: 1, _id: 0 })
db.flightsAIML64.find({ source: "Delhi" })
db.bookingsAIML64.find({ seats: 1 })
db.flightsAIML64.find().sort({ price: 1 })
db.bookingsAIML64.find().sort({ total_fare: -1 }).limit(2)
db.flightsAIML64.find({ price: { $gt: 5000 } })
db.flightsAIML64.find({ destination: { $ne: "Chennai" } })
db.bookingsAIML64.find({ seats: { $gte: 1, $lte: 2 } })
db.flightsAIML64.updateMany(
  { airline: "Indigo" },
  { $inc: { price: 500 } }
)
db.bookingsAIML64.deleteMany({ seats: { $gt: 2 } })
```

```
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{
    "acknowledged" : true,
    "insertedId" : null,
    "matchedCount" : 28.0,
    "modifiedCount" : 28.0,
    "upsertedCount" : 0.0
}
{
    "acknowledged" : true,
    "deletedCount" : 14.0
}
```

Q12. Find total fare collected city-wise. *(Aggregation)*

```mongo
use airlineDBAIML064
db.flightsAIML64.insertMany([
  { flight_id: 1, airline: "Indigo", source: "Delhi", destination: "Mumbai",
  { flight_id: 2, airline: "AirIndia", source: "Kolkata", destination: "Cher
  { flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Banga
  { flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chennai
  { flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Delhi'
  { flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "Kolk
])
db.bookingsAIML64.insertMany([
  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", s
  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata",
  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai",
  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", s
  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai",
  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore
])
db.flightsAIML64.find()
db.bookingsAIML64.find({}, { passenger_name: 1, city: 1, _id: 0 })
db.flightsAIML64.find({ source: "Delhi" })
db.bookingsAIML64.find({ seats: 1 })
db.flightsAIML64.find().sort({ price: 1 })
db.bookingsAIML64.find().sort({ total_fare: -1 }).limit(2)
db.flightsAIML64.find({ price: { $gt: 5000 } })
db.flightsAIML64.find({ destination: { $ne: "Chennai" } })
db.bookingsAIML64.find({ seats: { $gte: 1, $lte: 2 } })
db.flightsAIML64.updateMany(
  { airline: "Indigo" },
  { $inc: { price: 500 } }
)
db.bookingsAIML64.deleteMany({ seats: { $gt: 2 } })
db.bookingsAIML64.aggregate([
  { $group: { _id: "$city", total_fare: { $sum: "$total_fare" } } }
])
```

Find Query (li... | Shell Output (... | Aggregate Que...

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```
{
    "_id" : "Bangalore",
    "total_fare" : NumberInt(82500)
}
{
    "_id" : "Mumbai",
    "total_fare" : NumberInt(204000)
}
{
    "_id" : "Delhi",
    "total_fare" : NumberInt(75000)
}
{
    "_id" : "Kolkata",
    "total_fare" : NumberInt(195000)
}
```

Q13. Find average flight price per airline. *(Aggregation)*

```mongo
db.flightsAIML64.insertMany([
  { flight_id: 1, airline: "Indigo", source: "Delhi", destination: "Mumba
  { flight_id: 2, airline: "AirIndia", source: "Kolkata", destination: "C
  { flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Ba
  { flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chen
  { flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Del
  { flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "K
])
db.bookingsAIML64.insertMany([
  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi"
  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkat
  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai
  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi"
  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai
  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangal
])
db.flightsAIML64.find()
db.bookingsAIML64.find({}, { passenger_name: 1, city: 1, _id: 0 })
db.flightsAIML64.find({ source: "Delhi" })
db.bookingsAIML64.find({ seats: 1 })
db.flightsAIML64.find().sort({ price: 1 })
db.bookingsAIML64.find().sort({ total_fare: -1 }).limit(2)
db.flightsAIML64.find({ price: { $gt: 5000 } })
db.flightsAIML64.find({ destination: { $ne: "Chennai" } })
db.bookingsAIML64.find({ seats: { $gte: 1, $lte: 2 } })
db.flightsAIML64.updateMany(
  { airline: "Indigo" },
  { $inc: { price: 500 } }
)
db.bookingsAIML64.deleteMany({ seats: { $gt: 2 } })
db.bookingsAIML64.aggregate([
  { $group: { _id: "$city", total_fare: { $sum: "$total_fare" } } }
])
db.flightsAIML64.aggregate([
  { $group: { _id: "$airline", avg_price: { $avg: "$price" } } }
```

Shell Output (... | Aggregate Que...

50

```
{
    "_id" : "Vistara",
    "avg_price" : 7000.0
}
{
    "_id" : "SpiceJet",
    "avg_price" : 4000.0
}
{
    "_id" : "Indigo",
    "avg_price" : 7062.5
}
{
    "_id" : "AirIndia",
    "avg_price" : 6500.0
}
{
    "_id" : "GoFirst",
    "avg_price" : 4800.0
}
```

Q14. Display airlines where average price is greater than 5500. *(Aggregation + filter)*

```
  { flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Del
  { flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "K
])
db.bookingsAIML64.insertMany([
  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi",
  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkat
  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai
  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi",
  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai
  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangal
])
db.flightsAIML64.find()
db.bookingsAIML64.find({}, { passenger_name: 1, city: 1, _id: 0 })
db.flightsAIML64.find({ source: "Delhi" })
db.bookingsAIML64.find({ seats: 1 })
db.flightsAIML64.find().sort({ price: 1 })
db.bookingsAIML64.find().sort({ total_fare: -1 }).limit(2)
db.flightsAIML64.find({ price: { $gt: 5000 } })
db.flightsAIML64.find({ destination: { $ne: "Chennai" } })
db.bookingsAIML64.find({ seats: { $gte: 1, $lte: 2 } })
db.flightsAIML64.updateMany(
  { airline: "Indigo" },
  { $inc: { price: 500 } }
)
db.bookingsAIML64.deleteMany({ seats: { $gt: 2 } })
db.bookingsAIML64.aggregate([
  { $group: { _id: "$city", total_fare: { $sum: "$total_fare" } } }
])
db.flightsAIML64.aggregate([
  { $group: { _id: "$airline", avg_price: { $avg: "$price" } } }
])
db.flightsAIML64.aggregate([
  { $group: { _id: "$airline", avg_price: { $avg: "$price" } } },
  { $match: { avg_price: { $gt: 5500 } } }
])
```

```
Aggregate Que...   Aggregate Que...   Aggregate Q
50   Documents 1 to
{
    "_id" : "AirIndia",
    "avg_price" : 6500.0
}
{
    "_id" : "Indigo",
    "avg_price" : 7455.882352941177
}
{
    "_id" : "Vistara",
    "avg_price" : 7000.0
}
```

Q15. Find maximum total_fare among bookings for each city. *(Aggregation)*

```
db.bookingsAIML64.insertMany([
  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi",
  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkat
  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai
  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi",
  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai
  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangal
])
db.flightsAIML64.find()
db.bookingsAIML64.find({}, { passenger_name: 1, city: 1, _id: 0 })
db.flightsAIML64.find({ source: "Delhi" })
db.bookingsAIML64.find({ seats: 1 })
db.flightsAIML64.find().sort({ price: 1 })
db.bookingsAIML64.find().sort({ total_fare: -1 }).limit(2)
db.flightsAIML64.find({ price: { $gt: 5000 } })
db.flightsAIML64.find({ destination: { $ne: "Chennai" } })
db.bookingsAIML64.find({ seats: { $gte: 1, $lte: 2 } })
db.flightsAIML64.updateMany(
  { airline: "Indigo" },
  { $inc: { price: 500 } }
)
db.bookingsAIML64.deleteMany({ seats: { $gt: 2 } })
db.bookingsAIML64.aggregate([
  { $group: { _id: "$city", total_fare: { $sum: "$total_fare" } } }
])
db.flightsAIML64.aggregate([
  { $group: { _id: "$airline", avg_price: { $avg: "$price" } } }
])
db.flightsAIML64.aggregate([
  { $group: { _id: "$airline", avg_price: { $avg: "$price" } } },
  { $match: { avg_price: { $gt: 5500 } } }
])
db.bookingsAIML64.aggregate([
  { $group: { _id: "$city", max_fare: { $max: "$total_fare" } } }
])
```

```
Aggregate Que...   Aggregate Que...   Aggreg
50   Documents
{
    "_id" : "Kolkata",
    "max_fare" : NumberInt(13000)
}
{
    "_id" : "Delhi",
    "max_fare" : NumberInt(5000)
}
{
    "_id" : "Mumbai",
    "max_fare" : NumberInt(9600)
}
{
    "_id" : "Bangalore",
    "max_fare" : NumberInt(5500)
}
```